BIAS SIMILARITY MEASUREMENT: A BLACK-BOX AUDIT OF FAIRNESS ACROSS LLMS

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ABSTRACT

Conventional fairness evaluations grade large language models (LLMs) in isolation, masking how biases align, diverge, or persist across model families and releases. We introduce **Bias Similarity Measurement** (BSM), a unified black-box framework that reframes fairness as a *relational* property by comparing model bias signatures across scalar, distributional, behavioral, and representational views. BSM builds pairwise similarity profiles from six complementary metrics—accuracy, bias scores, output histograms, cosine distance, abstention (UNK) flip rates, and centered kernel alignment (CKA)—evaluated on bias-sensitive prompts drawn from BBQ, UnQover, and StereoSet. We conduct the largest bias similarity study to date, evaluating 30 LLMs spanning the LLaMA, Gemma, Gemini, and GPT families on over one million prompts. Our results reveal that instruction tuning primarily induces abstention rather than genuine representational debiasing: base–tuned CKA similarity remains above 0.94 (diagonal) and 0.85 (full), while models such as Gemma 2 9B-It and Gemma 3 12B-It flip more than 50% of biased answers to "Unknown." Critically, forced-choice evaluation (UnQover) exposes stereotypical leanings that abstention-based metrics conceal. We further show that small-model tuning can be regressive—LLaMA 3.2 3B worsens from s_{DIS}=17.67 to 30.97 under instruction tuning—while open-weight frontier models (Gemma 3 27B-It) achieve fairness competitive with proprietary counterparts. BSM provides practical workflows for procurement, release regression testing, and lineage screening, and demonstrates that fairness must be audited comparatively, not in isolation.

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1 INTRODUCTION

The proliferation of large language models (LLMs) across high-stakes domains—from hiring and healthcare to legal decision support—has made the evaluation of model fairness an urgent practical concern (Ferrara, 2023; Gallegos et al., 2024). A rapidly growing body of work has proposed benchmarks and metrics to probe stereotypical associations, question-answering biases, and demographic disparities in LLM outputs (Parrish et al., 2022; Li et al., 2020; Nadeem et al., 2021; Nangia et al., 2020). Despite this progress, existing approaches share a fundamental limitation: they evaluate models *in isolation*, reporting absolute bias scores that do not reveal how biases are inherited, evolved, or redistributed across model families and successive releases.

This isolation-centric view misses critical questions that arise in practice. When a vendor releases an instruction-tuned variant of a base model and claims improved fairness, has the model genuinely learned less biased representations, or has it merely learned to abstain more aggressively? When two models from different families achieve similar aggregate bias scores, do their errors align on the same demographic groups? When procurement officers choose between open-weight and proprietary systems, can they assess whether reported fairness gains are consistent across evaluation conditions?

We argue that fairness should be treated as a *relational* property—one that is meaningfully characterized not just by the score of an individual model, but by the *similarity structure* of bias signatures across a population of models. To this end, we introduce **Bias Similarity Measurement** (BSM), a black-box auditing framework that unifies multiple complementary bias signals into a single pairwise similarity space, enabling systematic comparison of models within and across families.

BSM operates on any LLM without access to weights or logits beyond token outputs, making it applicable to proprietary APIs as well as open-weight systems. It constructs bias signatures from six metrics spanning four views: (1) *scalar* metrics (accuracy and bias score), (2) *distributional* metrics (output histograms and Jensen-Shannon divergence (Lin, 1991)), (3) *behavioral* metrics (abstention flip rates under forced-choice reframing), and (4) *representational* metrics (CKA-based layer similarity (Kornblith et al., 2019)). Crucially, BSM separates ambiguous contexts—where abstention is the normatively correct response—from disambiguated contexts that require a definite answer, disentangling "fairness via caution" from "fairness via representation."

We apply BSM to 30 LLMs spanning the LLaMA (Touvron et al., 2023; Dubey et al., 2024; Meta AI, 2024), Gemma (Team et al., 2024a;b; 2025), Gemini (Google AI Developers, 2025), and GPT (Radford et al., 2019; OpenAI, 2024; 2025) families, evaluating each on BBQ (Parrish et al., 2022), UnQover (Li et al., 2020), and StereoSet (Nadeem et al., 2021)—over one million prompts in total. Our study yields four major findings:

1. Instruction tuning *primarily induces abstention*, not representational debiasing: base-to-tuned CKA similarity exceeds 0.94 on the diagonal and 0.85 over full layer matrices for all examined families.
2. Forced-choice evaluation *unmasks hidden bias*: models that appear nearly neutral under BBQ ambiguous conditions (with abstention available) reveal strong stereotypical leanings under the UnQover forced-choice protocol.
3. Small-model instruction tuning can be *regressive*: LLaMA 3.2 3B's disambiguated bias score worsens from 17.67 to 30.97 after tuning, while LLaMA 3.1 8B drops from 31.37 to 4.78.
4. Open-weight frontier models *can match or surpass* proprietary systems: Gemma 3 27B-It achieves near-neutral bias scores comparable to GPT-4o Mini, challenging the assumption that proprietary systems are inherently fairer.

These results demonstrate that BSM reveals patterns invisible to isolated evaluation and provides actionable information for model procurement, regression testing, and lineage auditing.

2 Related Work

Bias Benchmarks for LLMs. Several benchmarks have been proposed to quantify stereotypical associations and question-answering biases in language models. BBQ (Parrish et al., 2022) provides hand-built question sets in ambiguous and disambiguated settings across nine social dimensions, enabling separate assessment of bias under uncertainty and under clear evidential conditions. StereoSet (Nadeem et al., 2021) probes stereotypical associations through sentence completion tasks, while UnQover (Li et al., 2020) uses underspecified questions in a forced-choice format to reveal implicit stereotyping biases. CrowS-Pairs (Nangia et al., 2020) targets masked language models via minimal-pair sentence comparisons. More recent work has proposed compositional evaluation frameworks (Wang et al., 2025) and test suites covering ethical reasoning and demographic fairness (Abhishek et al., 2025). BSM builds on these benchmarks but takes a relational stance: rather than reporting absolute scores per model, it uses benchmark outputs as the raw material for cross-model similarity analysis.

Bias and Fairness Surveys. Gallegos et al. (2024) provide a comprehensive survey of bias evaluation and mitigation techniques, organizing metrics by embedding, probability, and text-generation levels. Ferrara (2023) surveys sources, impacts, and mitigation strategies for AI bias with particular attention to generative models. Kumar et al. (2024) conduct a large-scale study across 50+ LLMs finding that larger and newer models do not automatically exhibit reduced bias. Bai et al. (2024) introduce implicit bias measures that reveal latent stereotyping in explicitly value-aligned models, motivating our emphasis on behavioral (forced-choice) analysis alongside abstention-sensitive metrics. Morehouse et al. (2025) advocate for principled frameworks drawn from social science when designing bias probes, informing our separation of ambiguous from disambiguated evaluation contexts.

Model Similarity and Representational Analysis. Centered Kernel Alignment (CKA) (Kornblith et al., 2019) provides a principled measure of representational similarity between neural network layers, and has been applied to compare LLM internal representations across scales and training

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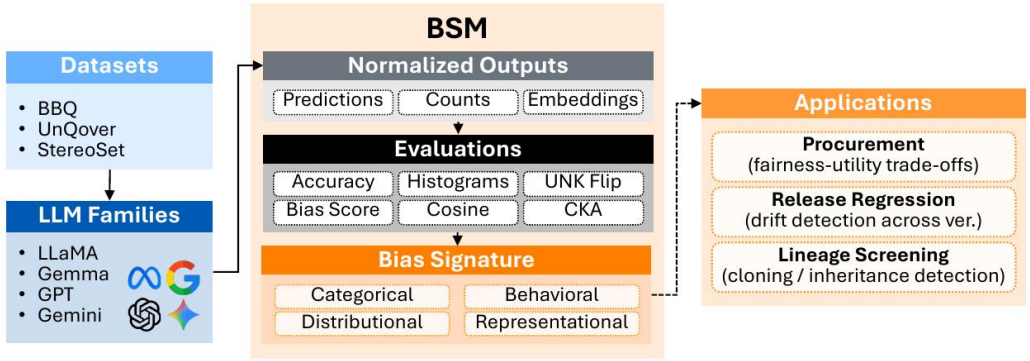

Figure 1: **BSM Pipeline.** Starting from bias-sensitive prompts drawn from BBQ, UnQover, and StereoSet, BSM elicits model outputs and computes six complementary bias metrics across four views (scalar, distributional, behavioral, representational). Pairwise similarity matrices are constructed within and across model families for procurement, regression, and lineage analysis.

stages (Klabunde et al., 2023; 2025). Li et al. (2021) propose behavioral similarity measures for model reuse detection. JSdivergence (Lin, 1991) and cosine similarity (Azarpanah & Farhadloo, 2021) have been used to compare output distributions and word embeddings across models. BSM synthesizes these techniques into a unified auditing pipeline specifically oriented toward fairness.

Instruction Tuning and Safety. Instruction-tuned and RLHF-aligned models often exhibit markedly different behavioral profiles from their base counterparts. Prior work has shown that tuning can reduce explicit stereotyping while preserving implicit associations (Bai et al., 2024), and that refusal-based safety mechanisms can mask rather than remedy underlying biases. The abstention flip rate introduced in BSM operationalizes this phenomenon directly, quantifying how much of an apparent bias reduction is attributable to increased "Unknown" responses rather than corrected stereotypical reasoning.

3 Bias Similarity Measurement

3.1 Overview

BSM is a black-box framework for comparing the fairness signatures of any pair of LLMs. Given a shared set of bias-sensitive prompts $\mathcal{P}$ and a set of models $\{M_1, \ldots, M_N\}$, BSM elicits responses from each model and constructs a six-dimensional bias signature $\mathbf{s}(M_i) \in \mathbb{R}^6$ per model, enabling pairwise similarity matrices within and across model families. Figure 1 illustrates the end-to-end pipeline.

3.2 Bias Metrics

BSM computes the following six metrics for each model on the shared prompt set:

(1) Accuracy (α). For QA benchmarks with a ground-truth answer, accuracy measures the fraction of questions answered correctly. BBQ disambiguated contexts provide clean answer keys; accuracy on these settings reflects both bias avoidance and factual reasoning ability.

(2) Bias Score (s). Following Parrish et al. (2022), we compute separate bias scores for ambiguous (s_{AMB}) and disambiguated (s_{DIS}) contexts. Positive values indicate stereotypical responding, negative values indicate anti-stereotypical responding, and values near zero indicate neutrality. Formally, for a set of questions $\mathcal{Q}$ with target groups g_1, g_2:

$$s = \frac{N_{\text{stereo}} - N_{\text{anti}}}{N_{\text{non-UNK}}} \times 100, \tag{1}$$

where N_{stereo} and N_{anti} count stereotypical and anti-stereotypical non-abstention responses, respectively, and $N_{\text{non-UNK}}$ excludes "Unknown" answers.

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(3) Output Histogram (H). For each model, we aggregate the distribution of responses across answer choices (target group 1, target group 2, Unknown) to produce a normalized histogram $H \in \Delta^2$. These histograms capture the full distributional profile of model behavior beyond scalar summaries.

(4) Cosine Distance ($d_{\cos}$). Pairwise cosine distance between output histograms measures how differently two models distribute responses across answer options, capturing systematic divergences in response tendencies.

(5) Abstention Flip Rate (f_{UNK}). The flip rate quantifies the fraction of prompts for which a model's answer changes from a group-targeted response to "Unknown" (or vice versa) when instruction tuning is applied. High flip rates indicate that tuning primarily modifies the abstention tendency rather than the underlying stereotypical direction of non-abstention responses.

(6) CKA Representational Similarity (ρ_{CKA}). For open-weight models, we compute centered kernel alignment (Kornblith et al., 2019) between corresponding layers of base and instruction-tuned variants using the linear kernel:

$$\rho_{\text{CKA}}(X, Y) = \frac{\|Y^\top X\|_F^2}{\|X^\top X\|_F \|Y^\top Y\|_F}, \tag{2}$$

where X and Y are layer activation matrices. We report both *diagonal* CKA (corresponding layer pairs) and *full* CKA (all layer-pair combinations).

3.3 Evaluation Contexts: Ambiguous vs. Disambiguated

A key design choice in BSM is the explicit separation of two prompt conditions that prior work often conflates:

- **Ambiguous (AMB):** The context does not provide enough information to determine the correct answer. The normatively appropriate response is abstention ("Unknown"). High accuracy and low abstention in this setting signal reliance on stereotypical heuristics.
- **Disambiguated (DIS):** The context provides sufficient evidence to determine the correct answer, which may or may not align with a stereotype. Bias in this setting reflects whether stereotypical associations override factual evidence.

This separation is essential for distinguishing "fairness via caution" (high abstention in AMB contexts) from "fairness via representation" (low stereotypical responding in DIS contexts). As we show in Section 5, instruction-tuned models frequently exhibit the former without the latter.

3.4 Similarity Space Construction

Given the six-dimensional bias signatures $\{\mathbf{s}(M_i)\}_{i=1}^N$, BSM constructs pairwise similarity matrices $\Sigma \in \mathbb{R}^{N \times N}$ for each metric dimension. These matrices reveal:

- **Family inheritance:** high within-family similarity on bias score and histogram metrics indicates shared training data or alignment procedures.
- **Cross-family convergence:** off-diagonal blocks with high similarity indicate convergence in bias profiles despite architectural differences.
- **Temporal drift:** comparing successive releases within a family (e.g., LLaMA 2 $\rightarrow$ 3 $\rightarrow$ 3.1) traces how fairness evolves across development cycles.

4 Evaluation Setup

4.1 Models

We evaluate 30 LLMs spanning four major families, including both base and instruction-tuned variants where available:

- **LLaMA family** (Touvron et al., 2023; Dubey et al., 2024; Meta AI, 2024): Vicuna (Chiang et al., 2023), Alpaca (Taori et al., 2023), LLaMA 2 7B (base + chat), LLaMA 3 8B and 70B (base + instruct), LLaMA 3.1 8B and 70B (base + instruct), LLaMA 3.2 3B (base + instruct).

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- **Gemma family** (Team et al., 2024a;b; 2025): Gemma 7B (base + instruct), Gemma 2 9B and 27B (base + instruct), Gemma 3 4B, 12B, and 27B (base + instruct).
- **Gemini family** (Google AI Developers, 2025): Gemini 1.5 Flash and Gemini 2.0 Flash (API, instruction-tuned only).
- **GPT family** (Radford et al., 2019; OpenAI, 2024; 2025): GPT-2 (legacy baseline), GPT-4o Mini, GPT-5 Mini (API).

4.2 DATASETS

BBQ (Parrish et al., 2022). We use the full BBQ benchmark covering nine bias categories (age, disability, gender, nationality, physical appearance, race/ethnicity, religion, SES, sexual orientation). Each category provides paired ambiguous and disambiguated question sets with three answer choices: two group-referencing options and an explicit "Unknown" abstention. We evaluate all models in a zero-shot multiple-choice format.

UnQover (Li et al., 2020). UnQover provides underspecified questions probing gender, nationality, ethnicity, and religion stereotypes. Unlike BBQ, UnQover is administered in a forced-choice format without an abstention option, compelling models to commit to group-attributing responses. This design makes UnQover particularly effective for revealing latent stereotyping that abstention strategies can conceal.

StereoSet (Nadeem et al., 2021). StereoSet evaluates models on sentence completion tasks in an open-ended generation setting, covering gender, race, religion, and profession. We use StereoSet to study output distributions and sentiment trends across demographic dimensions, complementing the QA-focused BBQ and UnQover analyses.

4.3 SCALE

In total, our evaluation spans over one million prompt evaluations across all models and datasets, making it the largest bias similarity study conducted to date. All evaluations are performed in a black-box, zero-shot setting without access to model weights for closed-source systems.

5 RESULTS

5.1 BIAS SCORES: INSTRUCTION TUNING EFFECTS

Table 1 reports average bias scores across ambiguous (s_{AMB}) and disambiguated (s_{DIS}) BBQ contexts for all base and tuned model pairs. Several patterns emerge clearly.

Table 1: **Average bias scores.** "–": anti-stereotypical, "+": stereotypical, and values near 0 = neutrality. Shown for ambiguous (s_AMB) and disambiguated (s_DIS) contexts.

Base Model	Avg. s_AMB		Avg. s_DIS	
	Base	Tuned	Base	Tuned
LLaMA 2 7B	5.45	4.30	7.50	6.65
LLaMA 3 8B	-4.78	-0.66	-8.72	-2.10
LLaMA 3 70B	-1.42	0.55	-4.51	2.50
LLaMA 3.1 8B	18.59	1.38	31.37	4.78
LLaMA 3.1 70B	0.42	-0.15	0.81	-1.44
LLaMA 3.2 3B	11.95	15.71	17.67	30.97
Gemma 7B	1.81	2.05	0.69	0.95
Gemma 2 9B	0.08	0.18	6.83	-2.02
Gemma 2 27B	6.95	0.51	14.31	-1.45
Gemma 3 4B	-3.89	5.83	2.69	8.62
Gemma 3 12B	4.36	0.15	6.12	-0.17
Gemma 3 27B	-1.25	0.07	-0.26	-1.50
Gemini 1.5	--	2.37	--	3.07
Gemini 2.0	--	-4.17	--	-5.54
GPT-2	72.43	--	96.19	--
GPT-4o Mini	--	0.47	--	2.66
GPT-5 Mini	--	0.21	--	1.10

Dramatic reductions in LLaMA 3.1 8B. LLaMA 3.1 8B exhibits the largest bias reduction upon instruction tuning of any model in our study: s_{AMB} falls from 18.59 to 1.38 and s_{DIS} falls from 31.37 to 4.78—reductions of more than 93% and 85%, respectively. As we show in Section 5.4, this reduction is substantially driven by increased abstention rather than corrected stereotypical reasoning in forced-choice settings.

Near-neutral proprietary models. GPT-4o Mini (s_{AMB}=0.47, s_{DIS}=2.66) and GPT-5 Mini (s_{AMB}=0.21, s_{DIS}=1.10) exhibit near-perfect neutrality on BBQ, representing the best scalar fairness performance in our study. LLaMA 3.1 70B and Gemma 3 27B-It achieve comparable neutrality among open-weight models.

Small-model regressions. LLaMA 3.2 3B shows a concerning regression under instruction tuning: s_{AMB} worsens from 11.95 to 15.71, and s_{DIS} worsens from 17.67 to 30.97. This finding indicates

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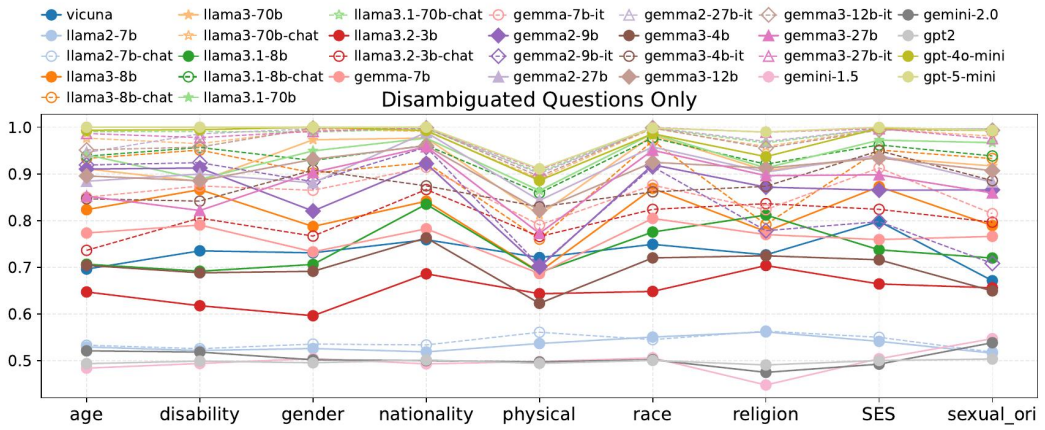

Figure 2: **Accuracy of LLMs on disambiguated BBQ questions.** Models are grouped by family and sorted within family. Instruction-tuned variants (shaded) generally improve over base models, but Gemini models and GPT-2 achieve very low accuracy despite differing bias profiles, reflecting high abstention rates (Gemini) or extreme stereotyping (GPT-2).

that instruction tuning is not uniformly beneficial and that smaller models may be more susceptible to tuning-induced bias amplification.

LLaMA 2 7B persistent stereotyping. LLaMA 2 7B remains stereotypical after chat tuning (s_{AMB}: 5.45 $\rightarrow$ 4.30; s_{DIS}: 7.50 $\rightarrow$ 6.65), with only marginal improvements, consistent with observations that early RLHF alignment addressed safety concerns more than demographic stereotyping.

Anti-stereotypical models. LLaMA 3 8B (base: s_{AMB}=-4.78) and Gemini 2.0 (s_{AMB}=-4.17) exhibit negative bias scores, indicating a tendency to respond anti-stereotypically. While anti-stereotypical responding is not itself harmful in the same way as stereotypical responding, it still reflects systematic deviation from group-neutral behavior.

5.2 Accuracy on Disambiguated Questions

Figure 2 shows accuracy on BBQ disambiguated questions across all models. Disambiguated accuracy measures whether models can leverage contextual evidence to override stereotypical priors—a different capability from bias score, which concerns non-abstaining responses only.

GPT-5 Mini achieves the highest accuracy across all models, followed by GPT-4o Mini, consistent with their near-zero bias scores. Among open-weight models, instruction-tuned LLaMA 3.1 70B and Gemma 3 12B/27B achieve competitive accuracy. Notably, Gemini 1.5 and 2.0 exhibit very low accuracy despite strong abstention tendencies, indicating that their apparent neutrality comes primarily from avoiding answers rather than answering correctly. GPT-2, by contrast, achieves low accuracy due to extreme stereotypical bias (Table 3).

5.3 Output Distributions: Gender Dimension

Figure 3 visualizes the output distribution histograms for the gender category, one of the most extensively evaluated bias dimensions. The histograms reveal stark differences between base and instruction-tuned variants: base models, particularly GPT-2, LLaMA 2 7B, and early Gemma variants, exhibit heavy skew toward stereotypically associated group answers, while instruction-tuned models—especially Gemma 2 9B-It and Gemma 3 12B-It—show dramatically inflated Unknown proportions, exceeding 50% of responses.

This distributional shift is the core empirical motivation for BSM's abstention flip rate metric: scalar bias scores improve dramatically with abstention inflation, but the underlying stereotypical distribution of *committed* responses may remain largely unchanged. The histogram view makes this pattern directly visible in a way that aggregate bias scores cannot.

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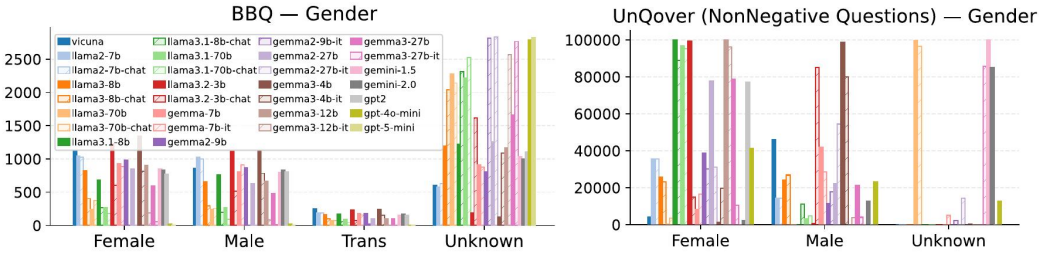

Figure 3: **Output distributions in the gender dimension.** Histograms show response proportions (male-associated, female-associated, Unknown) for each model on gender-probing BBQ prompts. Instruction-tuned models show dramatically increased Unknown rates, while base models exhibit marked stereotypical skews.

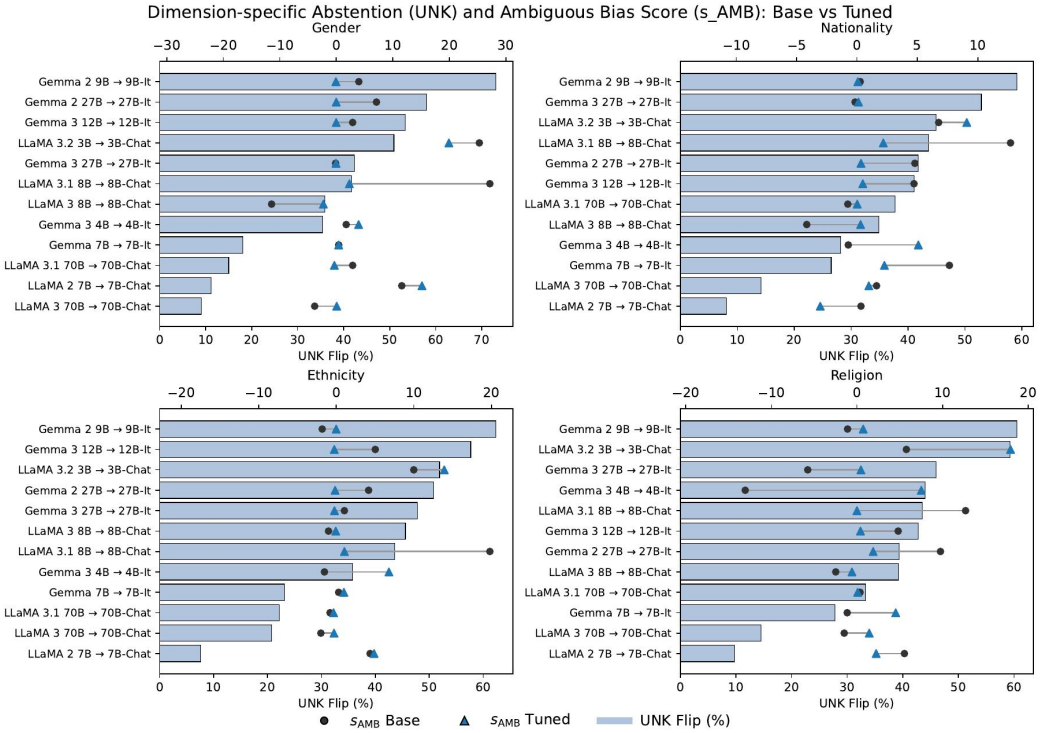

Figure 4: **Bias flip rates across models and datasets.** Each panel shows the fraction of prompts for which a model's answer flips to "Unknown" after instruction tuning (BBQ) or under forced-choice conditions (UnQover). High flip rates on BBQ (top) combined with persistent stereotyping on UnQover (bottom) indicate that instruction tuning primarily induces abstention rather than corrected reasoning.

5.4 Abstention Flip Rates and the Abstention Illusion

Figure 4 presents abstention flip rates across models, providing the clearest evidence of the "abstention illusion"—the phenomenon whereby instruction tuning dramatically improves bias scores by increasing abstention, without correcting the underlying stereotypical direction of committed responses.

Gemma 2 9B-It and Gemma 3 12B-It flip more than 50% of biased answers to "Unknown" under instruction tuning on BBQ ambiguous contexts. LLaMA 3.1 8B-Instruct flips approximately 40% of answers. When evaluated on UnQover—which forces a group-attributing commitment—these

Table 3: **Overall evaluation summary by model.** Qualitative synthesis of accuracy, abstention, bias direction, and representational similarity trends.

Model (abbrev.)	Key observations
Vicuna / Alpaca	Strongly anti-stereotypical; low accuracy; low abstention.
LLaMA 2 7B	Remain stereotypical after tuning; very low accuracy; weaker fairness.
LLaMA 3 8B	Anti-stereotypical lean; accuracy improves with tuning; moderate drift from LLaMA 2.
LLaMA 3.1 8B	Large bias drop after tuning; accuracy improves with tuning; high abstention.
LLaMA 3.1 70B	Near-neutral after tuning; high accuracy; high abstention even under forced choice.
LLaMA 3.2 3B	Strongly stereotypical; low accuracy; low abstention; weaker fairness vs 3.1 peers.
Gemma 2 9B	Stereotypical in base; abstention increases with tuning.
Gemma 3 4B	Slight stereotypical bias; accuracy competitive with LLaMA mid-size.
Gemma 3 12B / 27B	Near-neutral after tuning; high CKA similarity; fairness competitive with closed ones.
Gemini 1.5 / 2.0	Strong abstention; 2.0 skews anti-stereotypical; very low accuracy.
GPT-2	Extremely stereotypical bias; very low accuracy and fairness; serves as legacy baseline.
GPT-4o Mini	Near-zero bias; high accuracy; balanced abstention–fairness.
GPT-5 Mini	Near-perfect neutrality; highest accuracy; strongest stability across metrics.

same models exhibit substantially higher stereotypical responding than their BBQ scores suggest, revealing that the apparent debiasing is heavily abstention-mediated.

This finding has direct practical implications: procurement decisions made solely on the basis of BBQ scores for instruction-tuned models may overestimate genuine fairness improvements. BSM's forced-choice evaluation channel (UnQover) provides an essential complement that exposes the underlying bias structure.

5.5 Representational Similarity: CKA Analysis

Table 2 reports CKA similarity between base and instruction-tuned model pairs for five open-weight families. Two patterns stand out.

Table 2: Average CKA scores.

Model	Diag	Full
LLaMA 2 7B	0.991	0.902
LLaMA 3 8B	0.973	0.851
Gemma 1 7B	0.981	0.896
Gemma 2 9B	0.941	0.906
Gemma 3 12B	0.972	0.911

High base-tuned CKA across all families. Diagonal CKA (corresponding layer pairs) ranges from 0.941 (Gemma 2 9B) to 0.991 (LLaMA 2 7B), and full CKA (all layer combinations) ranges from 0.851 (LLaMA 3 8B) to 0.911 (Gemma 3 12B). These values indicate that instruction tuning leaves the internal geometry of representations largely unchanged, consistent with the view that RLHF and instruction fine-tuning primarily modify surface-level output behavior rather than deep representational structure.

Disconnect between representation and behavior. Despite high CKA similarity, the same model pairs show dramatically different bias scores and flip rates (Table 1, Figure 4). This disconnect confirms that the output-level behavioral changes induced by instruction tuning—including the abstention inflation documented above—are not driven by substantive changes in the internal representations that govern model reasoning. True representational debiasing, which would be reflected in lower CKA similarity, appears to be largely absent from current instruction tuning regimes.

5.6 Overall Model Summary

Table 3 synthesizes qualitative observations across accuracy, abstention, bias direction, and representational similarity for all model groups. Several high-level patterns merit emphasis:

- **Legacy models as baselines:** GPT-2 exhibits extreme stereotypical bias (s_{AMB}=72.43, s_{DIS}=96.19) and very low accuracy, serving as an important anchor for measuring progress across the LLM landscape.
- **Proprietary frontier models:** GPT-4o Mini and GPT-5 Mini achieve the best combination of near-zero bias and high accuracy. Among open-weight models, LLaMA 3.1 70B and Gemma 3 12B/27B (instruction-tuned) come closest to this profile.
- **Family-level trends:** The Gemma 3 series shows a clear progression toward neutrality with scale, with Gemma 3 27B-It reaching near-neutral bias scores competitive with closed-source models. The LLaMA 3.x series shows inconsistent scaling behavior: LLaMA 3.1 8B improves dramatically, while LLaMA 3.2 3B regresses.

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- **Gemini abstention behavior:** Gemini 1.5 and 2.0 exhibit strong abstention tendencies. Gemini 2.0 skews anti-stereotypical, while Gemini 1.5 is closer to neutral, but both achieve very low accuracy, suggesting abstention-dominated response strategies.

6 Discussion and Conclusion

The relational audit imperative. Our results confirm that isolated fairness scores are insufficient for rigorous model evaluation. BSM demonstrates that two models can achieve similar aggregate bias scores through fundamentally different mechanisms—one through genuine near-neutral responding, another through abstention inflation—with profoundly different implications for deployment. The similarity space constructed by BSM surfaces these distinctions and enables principled comparison across model families and releases.

Abstention as a fairness surrogate. Across nearly all instruction-tuned models in our study, the primary mechanism of apparent bias reduction is increased abstention rather than corrected stereotypical reasoning. High CKA similarity between base and tuned models confirms that this behavioral shift does not correspond to substantive changes in internal representations. This finding calls for a reassessment of instruction tuning as a fairness intervention: current alignment procedures appear to teach models *when not to answer* rather than *how to reason without stereotypical priors*. Evaluation frameworks that rely exclusively on settings where abstention is available will systematically overestimate the fairness benefits of instruction tuning.

Forced-choice evaluation as an essential complement. The UnQover forced-choice protocol reveals stereotypical leanings that BBQ ambiguous evaluations conceal behind abstention. We recommend that fairness audits routinely include forced-choice evaluation to complement abstention-permissive benchmarks, particularly when comparing instruction-tuned models where abstention inflation is prevalent.

Scale and family effects. Our results reveal that scaling is not uniformly beneficial for fairness: LLaMA 3.2 3B's instruction tuning actually amplifies bias, while LLaMA 3.1 8B achieves dramatic reductions. Within the Gemma 3 family, larger models achieve more consistent near-neutral behavior. These patterns suggest that the interaction between model scale, training data, and alignment procedure shapes fairness outcomes in ways that are not predictable from parameter count alone—motivating the kind of systematic cross-family comparison that BSM enables.

Open-weight models as viable alternatives. A practically important finding of this study is that open-weight frontier models—particularly Gemma 3 27B-It and LLaMA 3.1 70B-Instruct—achieve fairness profiles competitive with leading proprietary systems on our evaluation suite. This challenges assumptions that proprietary models are inherently superior in terms of fairness, and suggests that organizations with transparency and auditability requirements can consider open-weight models without sacrificing fairness quality.

Limitations and future work. BSM currently evaluates English-language bias across a set of Western-centric bias categories. Extending the framework to multilingual, cross-cultural, and domain-specific bias settings is an important direction for future work. Additionally, while CKA provides a strong indicator of representational stability, future work could explore probing-based analyses to more directly characterize which representations encode stereotypical associations. Finally, BSM's pairwise similarity matrices provide rich comparative information but do not yet support causal claims about bias inheritance mechanisms; developing causal models of bias propagation across LLM families remains an open challenge.

Conclusion. We introduced BSM, a unified black-box framework for auditing LLM fairness as a relational property. Applied to 30 models across four major families on over one million prompts, BSM reveals that instruction tuning primarily induces abstention rather than representational debiasing, that forced-choice evaluation unmasks bias hidden by abstention strategies, and that small-model tuning can be regressive. These findings motivate a shift from isolated fairness scoring toward comparative bias similarity analysis as the standard for LLM fairness auditing.

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